Recovering Sparse Neural Connectivity from Partial Measurements

A Covariance-Based Approach with Granger-Causality Refinement

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Problem: Partial Observability

Goal: Recover $N \times N$ connectivity W from partial measurements across sessions.

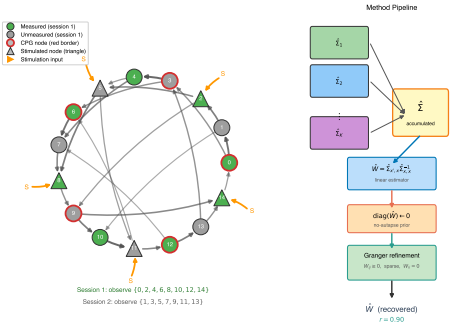
Dynamical model:

$$x_{t+1} = W\phi(x_t) + b_t$$

Session k observes $\mathcal{M}_k \subset \{1, \ldots, N\}$, stimulates $\mathcal{S}_k$.

Key insight: Pairwise covariance can be **accumulated across sessions** whenever i, j are co-observed.

C. elegans, zebrafish, organoids



Method: From Dynamics to Estimator

Step 1 — Covariance accumulation over $K{=}50$ sessions:

$$\hat{\Sigma}_{x_{t+1},x_t}(i,j) \;=\; \frac{1}{n_{ij}} \sum_{k:\, i,j \in \mathcal{M}_k} \hat{\Sigma}^{(k)}(i,j)$$

Step 2 — Linear estimator. Multiply $x_{t+1}{=}W\phi(x_t){+}b_t$ by $x_t^\top$, take expectations:

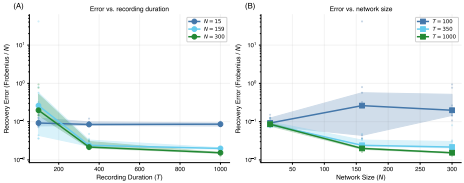
$$\Sigma_{x_{t+1},x_t} = W\,\Sigma_{\phi(x),x} + \Sigma_{b,x} \quad \xrightarrow{\phi(x)\approx x} \quad \boxed{\hat{W} = \hat{\Sigma}_{x_{t+1},x_t}\,\hat{\Sigma}^{-1}_{x_t,x_t}}$$

with diagonal zeroed ($W_{ii}{=}0$, no autapses) and non-negativity ($W_{ij}{\geq}0$).

Step 3 — Granger-causality refinement: Projected gradient descent enforcing sparsity zeros where contemporaneous covariance exceeds lagged.

Identifiability: $K \geq \log(N^2/\delta)/p^2$ sessions suffice for all-pairs coverage with probability $\geq 1{-}\delta$.

Result: $r = 0.90$ at 66% Coverage

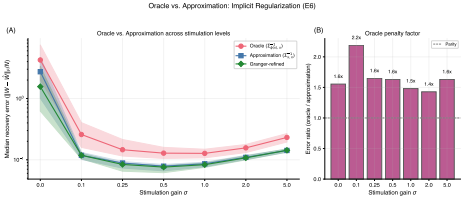


Error decreases with T and N.

Correlation	$r = 0.90$
Over same-prior	31% improvement
Best error	0.014 $(N{=}300, T{=}1000)$
Coverage	66% / session
Sessions	$K{=}50$

Granger refinement: precision $0.30 \rightarrow 0.40$, recall 0.97 — nearly all true edges preserved.

Surprise: The “Wrong” Model Wins



Oracle with known $\phi = \tanh$ is 1.5–2.7× worse.

We *deliberately* ignore the known nonlinearity.

Why? Stein–Price identity gives $\Sigma_{\phi(x),x} = D\Sigma_{x,x}$ where $D = \mathrm{diag}(\mathbb{E}[\mathrm{sech}^2(x_i)])$, $0 < d_i < 1$.

- Oracle inverts $D\Sigma$ — ill-conditioned
- Linear inverts Σ — biased but **better-conditioned**

$\Rightarrow$ **James–Stein shrinkage** in action.
No need to characterize the nonlinearity.

The Real Bottleneck: Intrinsic Dynamics

The estimation error decomposes into two terms:

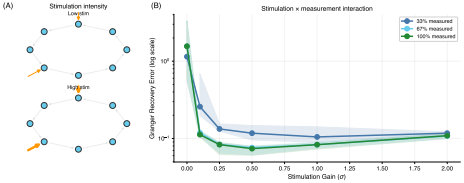
$$\underbrace{\hat{W} - W}_{\text{total}} = \underbrace{W(D - I)}_{E_1\text{: model mismatch}} + \underbrace{\Sigma_{b,x}\,\Sigma_{x,x}^{-1}}_{E_2\text{: CPG correlation}}$$

Error Source	$\|E\|_F/N$	**Relative**
E_1: Model mismatch (linear $\approx$ tanh)	0.080	$1\times$
E_2: CPG–state correlation	0.225	$\mathbf{2.8}\times$

Better recovery requires **modeling autonomous dynamics**, not refining the nonlinear approximation.

Practical Guidelines for Experimentalists

1. **Stimulate** moderately ($\sigma \approx 0.5$–1.0) on $\sim 33\%$ of neurons
2. **Cover** $\geq 50\%$ neurons per session
3. **Record** $K \geq 50$ sessions with random masks
4. **Use the linear estimator** — simpler *and* better
5. **Next:** Model CPG dynamics ($\sim 3\times$ dominant error)



The **control–estimation tradeoff**: stimulation aids identifiability but disrupts the signal.

Covariance accumulation turns partial observation from a limitation into a feature.

Paper & Code `github.com/qsimeon/sparse_matrix_recovery`

Contact `qsimeon@mit.edu`

Thank you!

Questions welcome at the poster session.